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Experimental investigation on solar PV panel dust cleaning with solution method

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ABSTRACT

The efficiency of solar PV panels varies depending on various factors; the type of material used to generate electrical energy, the quality of workmanship in the solar PV panel installation, environmental factors, dirt on the PV panel and design. Dust and dirt formed according to environmental conditions adhere to the solar PV panels and prevent the solar radiation from penetrating the surface. Thus, the solar PV panels need to be cleaned. In this study, three different chemical solutions prepared in laboratory conditions are applied to solar PV panels with a solar PV panel cleaning robot, which is manufactured using 3D printer technology to remove dust and dirt accumulated on solar PV panels for the first time in the literature. Thus, the effectiveness of chemical solutions to increase solar PV panel efficiency is demonstrated. The penetration of chemical solutions on the PV panel surface is ensured by the solar PV panel cleaning robot. The experimental set is realized under natural dust and dirt conditions. The effectiveness of the chemical solutions and electrical performance analysis results of solar PV panels are demonstrated by measurements and tests. The amount of power harvested from the PV panel cleaned using proposed Solution 1 (2-propanol) has been increased by 15%.

1. Introduction

The rapid increase in carbon emissions threatens the health and future of humans. Clean energy is obtained and energy demand is met thanks to energy systems based on renewable energy sources (Razmjoo et al., 2021; Elavarasan et al.,2020) Solar energy systems are one of the most preferred renewable energy sources in terms of their increased efficiency and gradually decreasing installation and maintenance costs (Kabir et al., 2018; Zhang et al., 2020). However, some key issues are encountered after the installation of solar energy systems, such as, the shading, dust and similar dirt types caused by different reasons cause a significant decrease in the energy obtained from solar energy systems (Madeti and Singh, 2017; Gupta et al., 2019; Mani and Pillai, 2010).

Many studies have been carried conducted in the literature on how to overcome dust and dirt difficulties for the PV array (Yilbas et al., 2021; Figgis and Bermudez, 2021; Derakhshandeh et al., 2021). The reports on energy efficiency losses in existing solar power plants in various countries of the world have been examined in detail. In addition, it has been comprehensively analyzed key information in solar power plant installation such as pollution losses in different locations and effective cleaning methods (Sayyah et al., 2014). Currently, various methods are used for cleaning PV panels, including cleaning by the classical method using a brush, removing dust from the surface with compressed air, coating technique and robotic cleaning systems. The dustiness level on PV panel surfaces in solar energy farms is quite high due to the high number of desert regions in the Middle East and African countries. Super-hydrophobic coating methods have been proposed to prevent PV panels from losing efficiency because of both organic and inorganic dirt accumulation. The super hydrophobic coating surface method for self-cleaning PV panels has many advantages such as anti-graffiti and anti-corrosion (Syafiq et al., 2018). The coating technique of PV panels with hydrophobic SiO_2 nanomaterial has been proposed to increase the amount of energy to be obtained from solar PV panels. The effect of the nanomaterial coating method on PV panel efficiency has been investigated experimentally in Assiut University Renewable Energy Laboratory in Egypt. According to the experimental results performed on three identical solar PV panels, it has been revealed that the implementation of the SiO_2 coating technique increases the efficiency of PV panels

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Table 1
The literature review on various cleaning methods of solar PV panels.

Ref. No.	Proposed Method	Advantages	Efficiency improvement
(Tejwani and Solanki, 2010)	360° Sun tracking-cum-cleaning system	Both sun tracking and automatically cleaning of the surface of solar panels; does not require any liquid for cleaning	Up to 30% more energy output as compared to the PV panel stationary on the ground and up to 15% more energy output as compared with single-axis tracking
(Al Shehri et al., 2016)	Dry cleaning using the robotic system	No need for cleaning liquid; does not cause any damage to the glass surface of the PV panel	No available data
(Assi et al., 2012)	Cleaning system based on return Air from Air Conditioning Systems	It prevents both dust accumulation and overheating of the Solar PV panel	Thermal energy on solar PV panels can be decreased reaching up to 73 %.
(Gheitasi et al., 2015)	Automatic cleaning system	Low-cost automation system; easy to implement for different power level PV panel array	The solar PV panel efficiency is maintained
(Khadka et al., 2020)	Robotic cleaning system	Cost-effective; provides remote control; can be integrated into a large-scale solar farm	No available data
(Lamont and El Chaar, 2011)	Microcontroller (PIC) and Programmable Logic Controller (PLC) based cleaning	Approximately 1.5 times more cost-effective than manual cleaning for offshore PV systems during a 10-year-period	No available data
(Odeh and Behnia 2009)	Solar-water pumping system	Very simple installation; low-cost cooling system; Efficient cleaning	Up to 15%
(Chaichan et al., 2015)	The cleaning method based on sodium surfactant	Low cost; eco-friendly; high efficient cleaning	Up to 15–16%
(Moharram et al., 2013)	Cleaning system using a mixture of anionic and cationic surfactants	Reduces the amount of water required for cleaning the Solar PV panel surface; can easily remove any dust that sticks to the surface	Efficiency remained constant thanks to the mixture of anionic and cationic surfactants
(Alqatari et al., 2015)	Three different cleaning systems are presented as air-blowing systems, superhydrophobic nano-coatings and electrodynamic screens (EDS).	Efficient cleaning systems; cost-effective; reduces annual maintenance costs	Up to 17.3% in EDS cleaning method; up to 16.8 % nano-coatings method; up to 10.2% Air-blowing cleaning system

(Alamri et al., 2020). Another cleaning approach is the electrostatic cleaning system, which uses a single-phase high voltage to remove sand from the solar panels' surface. It has been observed that the proposed cleaning method reduces the surface dirt of PV panels by 90%. The presented system has many advantages such as low cost and no need for consumables in solar power plants in desert regions at low latitude (Kawamoto and Shibata, 2015). There are roughly dry and aqueous cleaning approaches accessible in the literature on robot cleaning methods (Mondal and Bansal, 2015). A four-stage low-cost dry cleaning method is presented, which does not require any liquid for PV panel cleaning owing to water scarcity in these regions. The novel dry cleaning device that sprays jet air on the panel designed with high efficiency consists of a compressor, microcontroller, humidity sensor, light sensor rollers and motors (Aly et al., 2015). It is concluded that the increased harvest of solar energy by designing an automatic robotic dry cleaning system to minimize the dust on the surface of the solar PV panel. A new type of brush has been produced for the developed cleaning device, which is low cost and does not damage the PV panel surface (Parrott et al., 2018). The technical features of the automatic water-free cleaning device, which is designed indigenously to reduce dust and dirt on PV panels, are presented. The proposed water-free cleaning device has been observed by experimental studies carried out for one month, increasing the amount of energy obtained from PV panels by approximately 6.31% (Deb and Brahmbhatt, 2018). The dust absorbing flow rate of autonomous robots designed to increase the efficiency of PV panels directly affects the cleaning on panels surface. The mechanical behavior and pressure distribution of the dust particles in the dust suction process have been simulated based on the Computational Fluid Dynamics (CFD) software. The optimum dimensions for the suction inlet of the autonomous robot have been determined by the orthogonal experiment method (Cai et al., 2019). Simulation studies are performed using the Virtual Robot Experiment Platform to clean the dirt or dust on the surface of the PV panels with autonomous vehicles. The novel algorithms have been developed using the Robot Operating System to control the autonomous PV panel cleaning vehicle (Memon, 2016). A cleaning system that sprays water on the PV cells is designed to increase the efficiency of the PV water pumping system. The efficiency of the system has been significantly improved by reducing the high temperature in the PV cells with a water-spraying developed cleaning system (Abdolzadeh and Ameri, 2009). The literature review on various cleaning methods of solar PV panels is given in Table 1. Currently, various methods are used for cleaning PV panels, including cleaning by the classical method using a brush, removing dust from the surface with compressed air, natural cleaning due to precipitation, and robotic cleaning systems. There are roughly dry and aqueous cleaning approaches accessible in the literature on robot cleaning methods (Mondal and Bansal, 2015). In dry cleaning, the method of cleaning with a brush and absorbing dust was used, and an increase in the power production of about 9% was achieved (Deb and Brahmbhatt, 2018). In the aqueous method, it is the method used only by pouring water onto the surface of the PV panel. No significant increase in PV panel efficiency was observed in this method (Moharram et al., 2013). As these results show, using only non-pressurized water for cleaning PV panels is not an efficient method. Instead, it is much more effective to use pressurized water or to clean the PV panel surface by brushing it.

In this paper, a solar PV panel surface cleaning technique based on chemical solutions is proposed to maximize the amount of PV energy harvesting. Given the lack of use of the solutions presented in existing PV panel surface cleaning methods, the main contributions of this article can be summarized as follows:

- The surface cleaning solutions have been developed that is much more effective than water and natural cleaning for PV panel surface cleaning.
- The amount of PV energy harvesting has been maximized by using effective chemical solutions for PV panel surface cleaning. For this purpose, 2-propanol (5% v/v), ethanol (5% v/v) and acetone (5% v/v) solutions have been used Solution1, 2 and 3, respectively.
- The design and manufacture of the cleaning device called the solar PV panel cleaning robot have been conducted.
- Thanks to the special design of the PV cleaning robot, its integration on large-scale PV the array will be realized easily.

In the following sections of the paper, the electrical properties of used PV panel, developed measurement station, designed solar PV panel

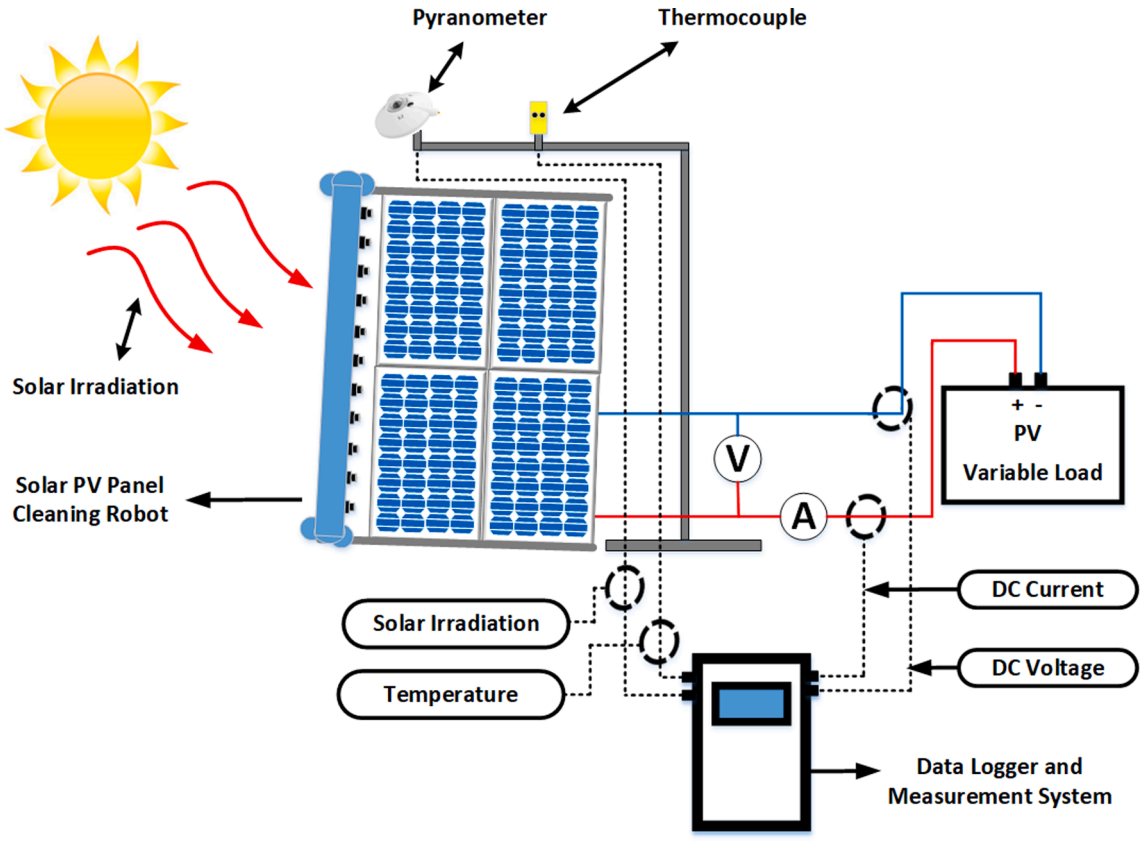


Fig. 1. Solar PV panel cleaning test system.

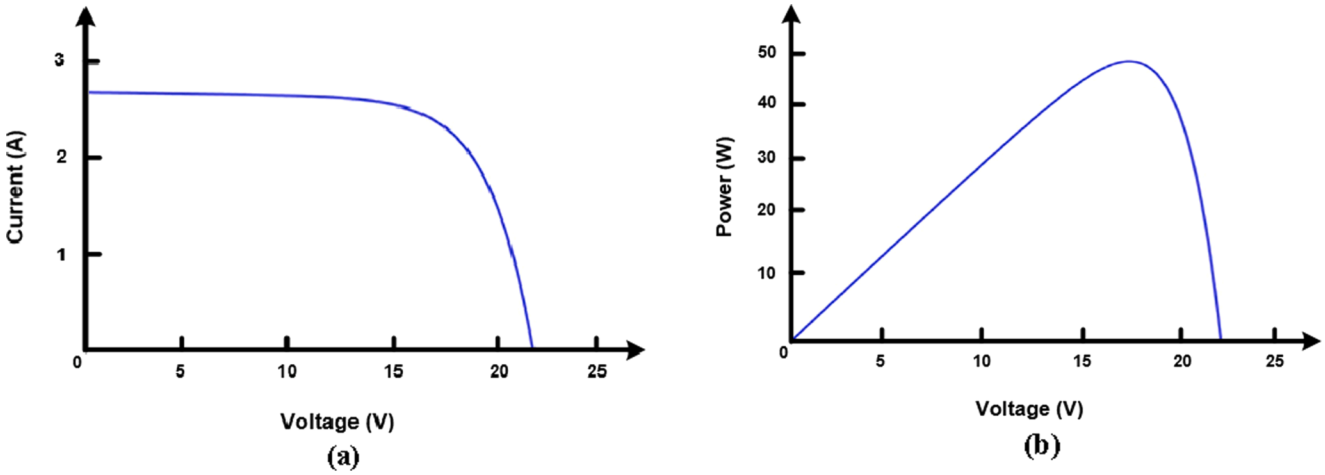


Fig. 2. Solar PV panel used in experimental study a) I-V characteristic b) P-V characteristic.

cleaning robot, chemical solutions experimental investigation, result and discussion have been presented comprehensively.

2. Materials and methods

Cleaning the solar energy-producing panels is important in terms of maintaining efficiency. Pure water and some alcohol cleaners are used for panel cleaning. Alcohol-based cleaners are suggested to avoid solar panels freezing owing to the effects of cold, especially in the winter. In this study, alcohol-based cleaners are preferred to prevent dust on solar panels. The effectiveness of the prepared solutions in dissolving the dust accumulated on the solar PV panels is investigated. For this reason, an experimental set including solar PV panel, data logger and measurement system, nozzle and chemical solutions is set up. The experimental test setup is given in Fig. 1. The solutions and water used to clean the panel surface have been not applied consecutively on the same panel. Experimental studies have been carried out on 4 separate 50 W PV panels dusted in the same environment. The surface of each panel is cleaned with different liquids (Solution 1, 2, 3 and water) and the short-circuit current and open-circuit voltage are measured simultaneously. In the experimental studies, one liter of each solution is used for one minute for each PV panel. The same amount of time has been spent spraying with the cleaning robot each solution. Experimental studies have been performed at least five times at different times to determine the order of

Table 2
The electrical and mechanical specifications of the solar PV panel used in the experimental study.

Electrical and Mechanical Characteristics	Value	Electrical and Mechanical Characteristics	Value
Power Rating	50 W	The voltage at maximum power (V_{mp})	17.3 V
Short Circuit Current (I_{sc})	2.95 A	Power Tolerances	±5%
Open Circuit Voltage (V_{oc})	21.6 V	Number of Cells	36
Maximum System Voltage	600 V	Current at maximum power (I_{mp})	2.76 A
Dimensions	470x67x22 mm	Weight	4,5 kg
Frame	Anodized Aluminum	Glass	3,2 mm, Tempered Glass

superiority of the solutions in cleaning the PV panel surface.

2.1. Solar PV panel

Solar PV panels formed by connecting a solar cell with serial or parallel connection of these cells are composed of a current source electrically, serial/parallel resistors and parallel diodes. The relationship between the voltage of solar cell cells and the current it gives to the load gives the I-V and P-V characteristics of the cell. These two characteristics provide important clues about the conditions under which the power received from the panel should reach its highest value. In this study, the performance of the solar PV panel used for the experimental study is evaluated with the I-V and P-V curves given in Fig. 2. Also, Table 2 presents the parameters of the solar PV panel used for the experimental study.

2.2. Data logger and measurement system

By using a data logger and measuring station, the solar radiation intensity reflected on the solar PV panel, the ambient temperature of the PV panel, the PV panel DC output current and voltage values were measured using a pyranometer, j type 4–20 mA temperature sensors, DC current and DC voltage sensors, respectively. Wind speed, current, voltage and solar radiation intensity data are taken with the help of the relevant sensors and sent to the addresses on the analog input modules. The data obtained in the CPU, which is communicated with analog input modules, is processed. The values taken from the sensors are output as current values between 4 mA and 20 mA and are scaled according to the minimum and maximum values of the sensors. The current rates that come with the software in the CPU are converted to the operating rates of the sensors and output is given. These output values are recorded with the software in the CPU and sent to the Excel file and saved to the Excel file with the desired time intervals. Thanks to the PLC measuring station, the data can be read instantly thanks to the HMI panel. PLC measuring station is as seen in Fig. 3.

2.3. Fogging nozzle

One of the most important components of a cooling system is the fogging nozzle. All fogging systems consist of a series of nozzles arranged in a line. The fogging nozzle is used for outdoor cooling in all kinds of industries and for all kinds of purposes. They are used to ensure

Table 3
The fogging nozzles used in the solar PV panel cleaning robot.

Fogging Nozzle	Type	Ejection Pressure (bar)	Ejection Angle (L/min)	Spray Radius (meter)	Spray Characteristics
Spray Sprinkler Head Irrigation Cooling	Single-Fluid	2–25	1.4–7.8	0.7–0.9	Small flow rate, very fine spray, easy to install and convenient to use.

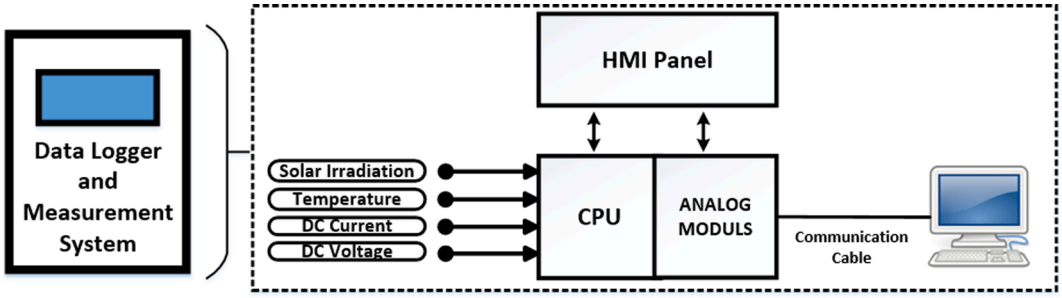


Fig. 3. Data logger and measurement system.

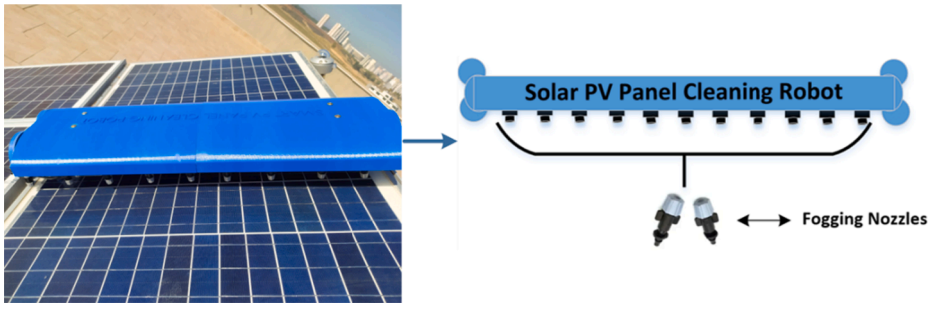


Fig. 4. Fogging nozzles used in the solar PV panel cleaning robot.



Fig. 5. Experimental set-up unit for 4x50 W solar PV panel.

the comfort of visitors in public spaces and commercial establishments and are also used in industrial facilities and agricultural applications. The fogging nozzles used in the solar PV panel cleaning robot are shown in Fig. 4.

Also, Table 3 shows the characteristics and performance data of the fogging nozzles used in the solar PV panel cleaning robot.

When the high-pressure fogging pump is connected, water is forced through the fogging nozzles and the mist particles evaporate when they reach the outside air. This can reduce temperatures by 10–15°. If the fogging system is used outdoors, it will create a cool area. if located in a confined space, it will provide a cool-mist but will also increase humidity.

2.4. Chemical solutions

Many organic solutions are recognized to use for surface cleaning. However, the cleaning solutions for PV panels should be determined specifically. In this study, the volatile solutions were preferred due to high solubility in water, available price, abundance and lower surface tension properties. When the surface tension of water is lowered, water molecules moisten the material more easily and penetrate the dirt quickly (Dufour et al., 2013). In this regard, 2-propanol (5% v/v), ethanol (5% v/v) and acetone (5% v/v) aqueous solutions were used and symbolized as Solution1, 2 and 3, respectively.

2.5. Surface characterization

The contact angles of solutions on PV panel glass were investigated by Biolin Scientific Attension Theta Lite Optical Tensiometer. The dust

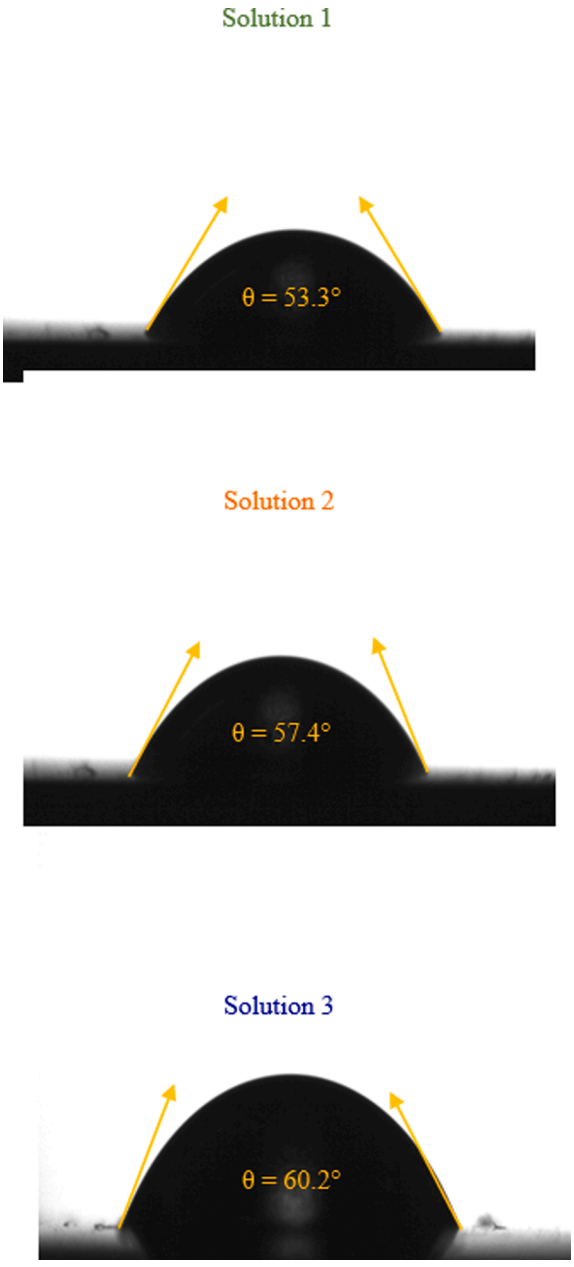


Fig. 6. The contact angle for solutions droplet on PV panel glass.

on the surface of PV panel was investigated by using scanning electron microscope by Carl Zeiss Evo 40 SEM equipment. The chemical composition was determined by EDX analysis with SEM.

3. Results and discussions

Experimental tests have been conducted to investigate the performance of four 50 W PV panels installed on the roof of Alparslan Turkes Science and Technology University in Adana under-five different cleaning conditions. The effect of dust on the short circuit current and open-circuit voltage of PV panels has also been recorded and experimentally observed using the PLC measurement station. The relationship between the solar radiation level and the produced power has been

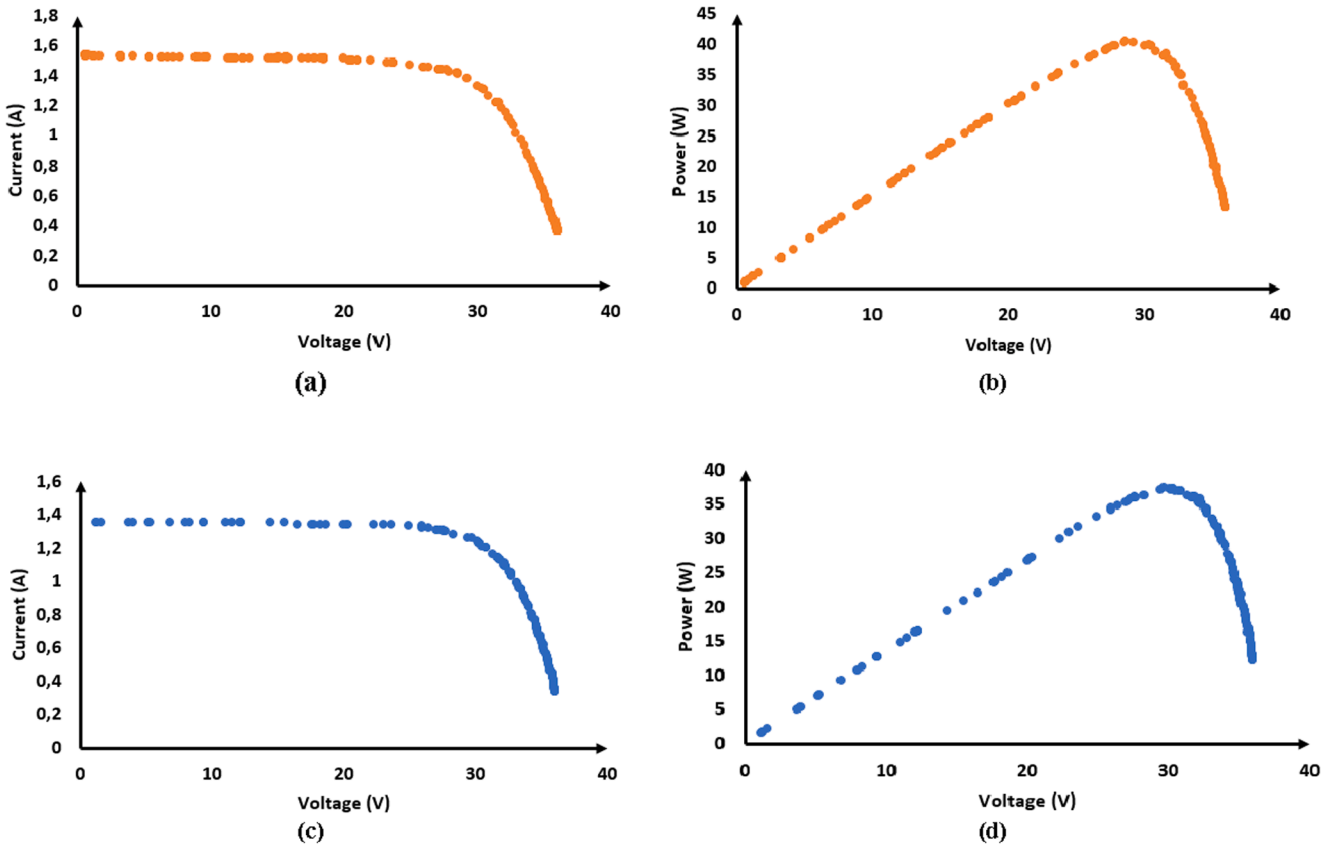


Fig. 7. Clean case solar PV Panel electrical characteristics a) I-V b) P-V; Dusty case solar PV Panel electrical characteristics c) I-V d) P-V.

observed by measuring the solar radiation by the pyranometer. The experimental set-up unit is given in Fig. 5.

Dusty photovoltaic panels were cleaned with a photovoltaic panel cleaning robot with pressurized water spraying fogging nozzles. Cleaning was carried out using water and 3 different solutions. The obtained contact angle values were 53.3°, 57.4° and 60.2° for Solution1, 2 and 3, respectively. As can be seen in Fig. 6, the values were less than 90, indicating that the solutions' wettability properties were successful. The I-V and P-V characteristics of clean solar PV panels and dusty solar PV panels are given in Fig. 7.

As seen from Fig. 8, the particle sizes of dust varied between almost 29 and 46 μm, also lower particles were detected with 10 μm diameters. The Si, O, Al, Na, S, K, Fe, Ca, etc. were detected in EDX spectra. The mapping analysis proved that Si, O and Al which were the main components, were homogenously dispersed on the dust surface, the values were 16.9; 57.2 and 10.3%, respectively (Fig. 8).

The dusty solar PV panel is indicated in Case 1. The PV panel, which is cleaned with water using the cleaning robot, is indicated as Case 2. The difference in power produced between Case 1 and Case 2 appears to be a difference of 3.7 W. The panel cleaned with water shows that 10% more power is produced than the dusty PV panel.

Solar PV panel temperature directly affects the power produced from the solar panels. Experimental studies are repeated to show that the solutions maintain their performance efficiency even with the solar PV panel temperature and solar radiation change. To prove this case, the relative maximum power points of the solar PV panels are examined that they stay constant for hours after the cleaning is performed. Table 5 presents that the measured power values from solar PV panels to which Solution 1, Solution 2, Solution 3 and water are applied, respectively. According to the experimental results, the superiority of Solution 1 over other solutions is observed again. The values at the maximum power point of the solar PV panels to which Solution 1, Solution 2, Solution 3 and water are applied are 49.92 W, 46.63 W, 43.64 W and 40.18 W respectively. Table 4 and Table 5 are the performance analysis results in different day after the solutions are applied on the solar PV panel surface.

After the temperatures of the PV panels are stabilized, the maximum power point of the PV panels is measured again to prove the effectiveness of the proposed solution. After the experimental validation analysis process in Table 5, it is waited for 2 h to receive experimental data from the solar PV panels again. Experimental analysis data of the solar PV after 2 h is given in Table 6. As a result of the comparison of data in Table 4, Table 5 and Table 6, it has revealed that the proposed Solution 1 maintains its superiority over the other solutions. The relative maximum power point of the panels stayed constant for hours after the cleaning is performed.

Different solutions have been tried to further increase the amount of generating power. In Table 4, the short-circuit current, open-circuit voltage, maximum power and ambient temperature values obtained as a result of different solution cleaning for each 50 W PV panel are shown comparatively. Solution 1 was sprayed under pressure with the help of a cleaning robot on the dusty panel and the panel was cleaned (Case 3). Measurements were performed after the solutions on the PV panel evaporated and dried. Case 3 has become 15% more efficient than Case 1

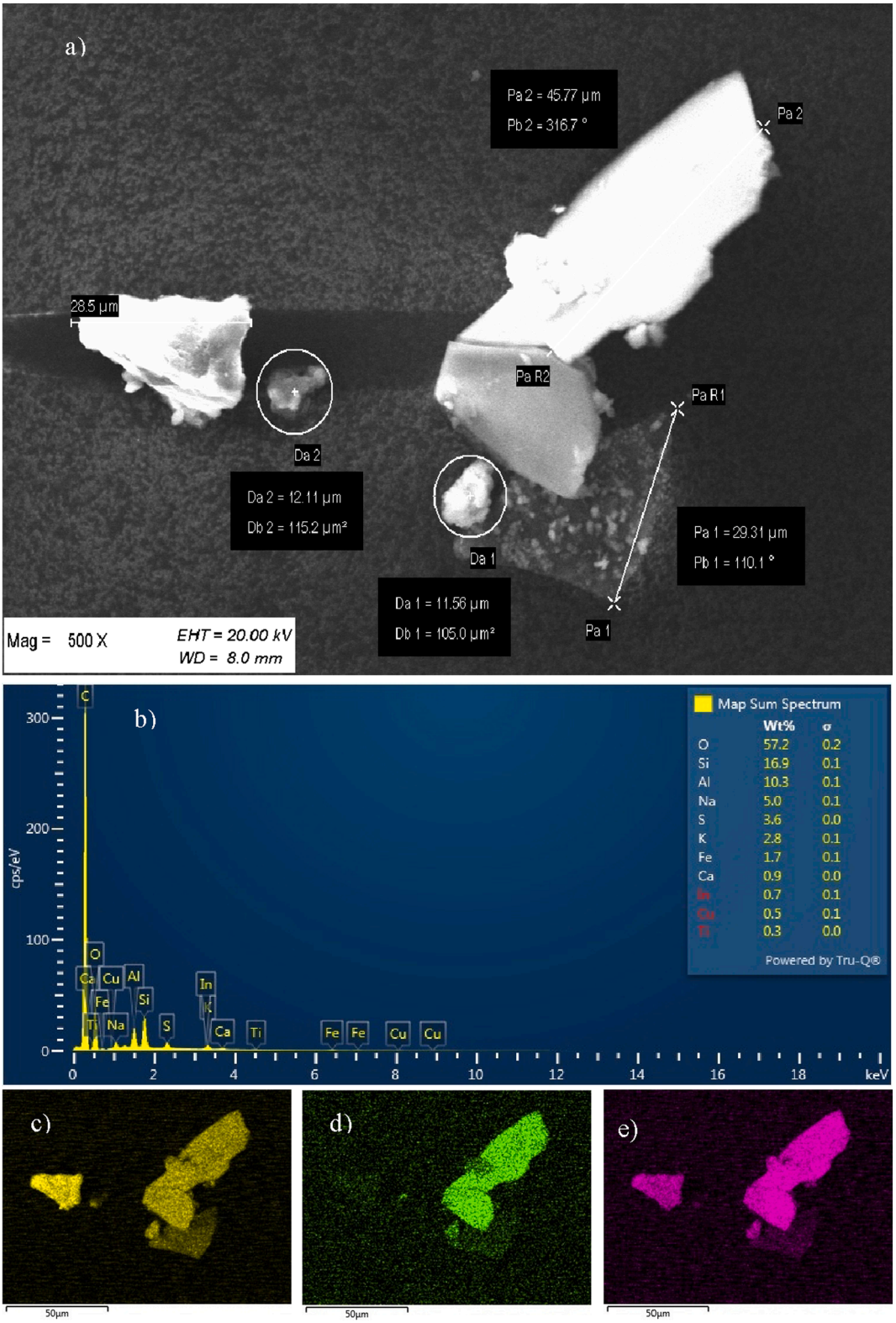


Fig. 8. The SEM micrograph of dust (a), the EDX analysis of dust (b), the EDX dot mapping of dust: Si yellow zones (c), O green zones (d) and Al pink zones (e). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Experimental analysis data of the solar PV panels at cleaning conditions.

Cases	PVs Cleaning Condition	P_{max} (W)	V_{max} (V)	I_{max} (A)	Ambient Temperature	Solar Radiation
Case 1	Dusty	36.84	33.4	1.25	28℃	912 W/m^2
Case 2	Water	40.54	28.6	1.41	28℃	912 W/m^2
Case 3	Solution 1	42.46	29.90	1.42	28℃	912 W/m^2
Case 4	Solution 2	41.98	29.70	1.41	28℃	912 W/m^2
Case 5	Solution 3	40.96	30.80	1.33	28℃	912 W/m^2

Table 5
Experimental validation analysis data of the solar PV panels in different day.

PVs Cleaning Condition	P_{max} (W)	Average Solar PV Panel Temperature	Average Solar Radiation
Water	40.18	46.8℃	1180 W/m^2
Solution 1	49.92	48.1℃	1180 W/m^2
Solution 2	46.63	48.2℃	1180 W/m^2
Solution 3	43.64	48.1℃	1180 W/m^2

Table 6
Experimental analysis data of the solar PV after 2 h.

PVs Cleaning Condition	P_{max} (W)	Average Solar PV Panel Temperature	Average Solar Radiation
Water	37.42	40.2℃	909 W/m^2
Solution 1	42.45	40.1℃	909 W/m^2
Solution 2	41.62	40.1℃	909 W/m^2
Solution 3	39.44	40.1℃	909 W/m^2

and 5% better than Case 2. Solution 2 and Solution 3 were used in Case 4 and Case 5 measurements, respectively. Case 4 and Case 5 achieved good results as a contribution to power generation compared to Case 1 and Case 2. Case 4 and Case 5 performed 14% and 11% better than Case 1 the dusty condition as shown in Fig. 9. P-V characteristics of solar PV panels in all cases obtained from experimental studies are given Fig. 10.

4. Conclusion

A solar PV panel surface cleaning technique based on chemical solutions was proposed to maximize the amount of PV energy efficiency. Therefore, an experimental set including solar PV panel, data logger and measurement system, nozzle and chemical solutions is set up. Experimental studies have been performed to investigate the performance of four 50 W PV panels under five different cleaning cases. The comparison of the aqueous solutions which were presented from Case 2 to Case 5 was proved that the PV performance was enhanced by using solutions. As a result, the wetting characteristics of these low surface tension liquids (Solutions from 1 to 3) are significantly altered, as they tend to wet the majority of surfaces. The amount of power harvested from the cleaned PV panel using Solution 1, Solution 2 and Solution 3 have been increased by 15%, 14% and 11%, respectively. The most convenient aqueous solution is Solution 1 (Case 3) due to effective solution properties and lower surface tension.

In future studies, effective aqueous solutions will be investigated not only for natural dust but also for other dirty conditions on solar PV panels. It is planned to develop the proposed solar PV panel cleaning robot and make it autonomous.

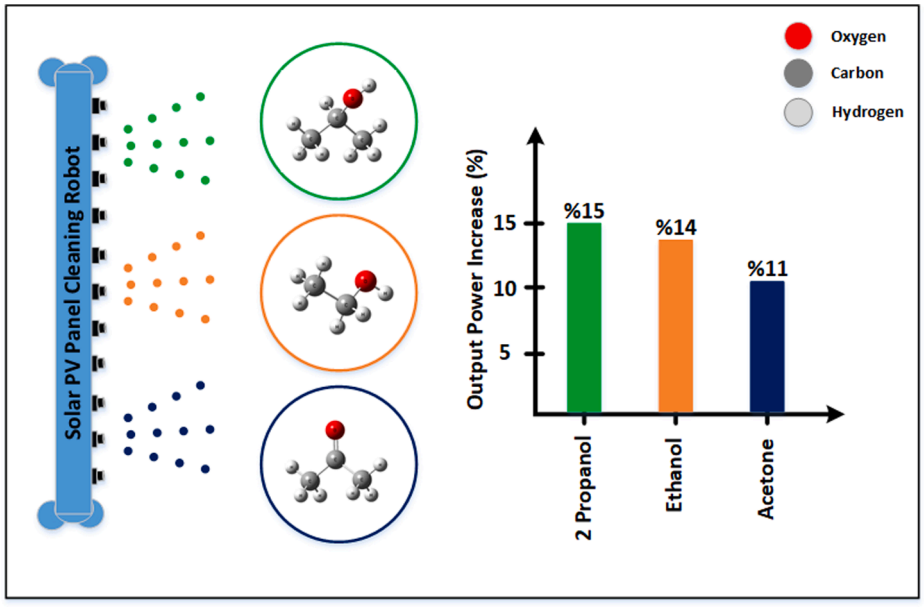


Fig. 9. Solution performances of solar PV panel efficiency increase.

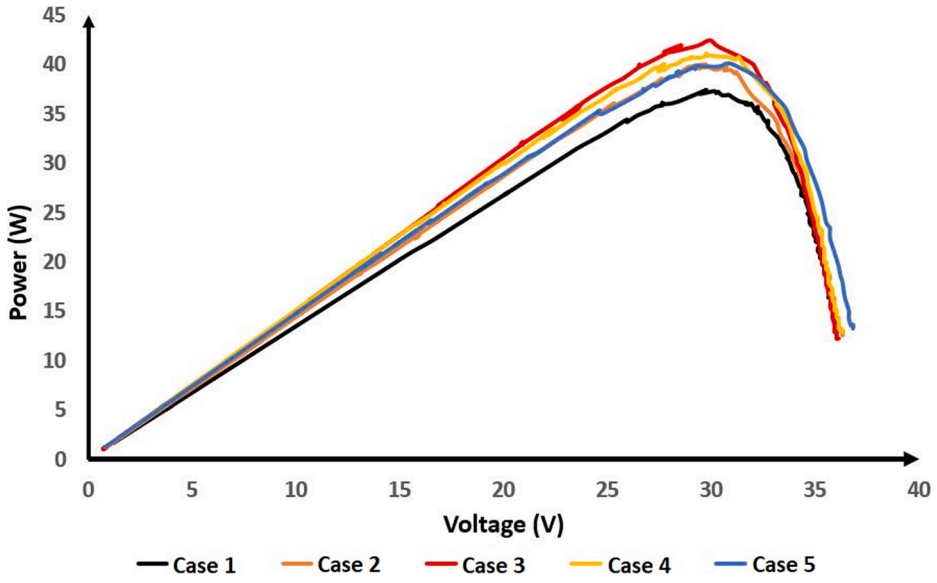


Fig. 10. P-V characteristics of solar PV panel in all cases.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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